

International Interactions

ISSN: 0305-0629 (Print) 1547-7444 (Online) Journal homepage: http://www.tandfonline.com/loi/gini20

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Alastair Smith

To cite this article: Alastair Smith (1998) Extended deterrence and alliance formation, International Interactions, 24:4, 315-343, DOI: 10.1080/03050629808434934

To link to this article: http://dx.doi.org/10.1080/03050629808434934

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International Interactions
Vol. 24, No. 4, pp. 315–343
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Printed in India.

EXTENDED DETERRENCE AND ALLIANCE FORMATION

ALASTAIR SMITH

Department of Political Science
Campus Box 1063
Washington University
One Brookings Drive
St. Louis, MO 63130

(Received for Publication April 21, 1997)

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Inductive approaches have little conclusive to say about whether alliances make war more or less likely. Moreover, the empirical associations are ambiguous. Modeling alliance formation in an incomplete information setting reveals that alliances have predictable effects on the behavior of nations. The primary effect of alliances is to increase the probability that a nation will intervene in an ongoing war. Since an alliance provides a signal that a nation is likely to intervene, alliances affect the behavior of other states. Specifically, the formation of a defensive alliance deters potential aggressors from attacking, and encourages those nations that are attacked to resist. Although an alliance helps deter enemies and helps reassure friends, success is not guaranteed; deterrence failures occur. Although alliances have a predictable effect on the behavior of nations, the aggregate relationship between alliance formation and the occurrence of war is ambiguous. As well as providing an explanation of much of the empirical evidence, this model generates new testable hypotheses.

KEY WORDS: alliance, extended deterrence, deterrence, war, conflict

Using a game theoretic framework, I model how alliances affect the interactions between nations. I examine a simple three nations system. Suppose that one of these nations, which I call nation A, is contemplating an attack on another nation, nation B. If nation A attacks, then nation B must decide whether to resist. If nation B decides to retaliate then A and B are at war. Once a war starts, the third nation, C, has the option of remaining neutral

or intervening in support of B. This paper examines how alliances affect the probability that A attacks (α), the probability that B resists (β), and the probability that C intervenes (γ).

The model I offer answers several questions: When do nations form alliances? Are alliances reliable? How does an alliance affect the behavior of nations A and B, and hence how are alliances related to the occurrence of war? I concentrate on defensive alliances between nations B and C. By offering nation B an alliance, nation C can potentially deter A from attacking. This is a situation of extended deterrence (Huth, 1988; Wu, 1990). The extended deterrence literature would describe nation A as a "potential attacker", nation B as a "protege" and nation C as a "defender". As well as examining alliance formation, this paper provides a model of deterrence.

Alliances are non-binding agreements between nations. Nations pledge to offer military assistance should an ally become involved in a war. In the anarchic system of international relations, alliances can never be binding. However, there are costs associated with the formation and the failure of alliances. These costs could be generated by reputation (Synder and Diesing, 1977; Alt, Calvert and Humes, 1988), by domestic politics (Morrow, 1991; Bueno de Mesquita *et al.*, 1992), or by loss of autonomy (Altfeld, 1988). For the purposes of this paper, the manner in which these costs are produced is irrelevant. What is important is that nations pay costs for forming alliances, and nations pay costs for not honoring their commitments.

The model considers formal alliances. However, the results of the model hold for any commitment that is costly to make or costly to break. Any form of costly pledge affects behavior in accordance with the predictions of the model. However, the magnitude of the effects vary with the cost of the commitment. Thus, as well as examining alliance formation, this paper models extended deterrence more generally. The rational choice approach to explaining deterrence has been much debated by both its proponents and its critics (Achen and Snidal, 1990; Downs, 1990; George and Smoke, 1990; Huth and Russett, 1990; Jervis, 1990; Lebow and Stein, 1990; Zagare, 1990). I choose rational choice over other approaches as it enables generalizable statements necessary for theory building to be constructed (Downs, 1990).

A theory of alliance behavior needs to explain why some nations form alliances and why others don't. It should also explain why some alliances are reliable and others are not. These seem fairly straightforward questions, but despite the large range of empirical research, the inductive approach has little conclusive to say about how alliances are associated with war. Singer and Small (1967, 1968) were among the first to empirically study

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the correlation between the formation of alliances and the occurrence of war. Despite numerous subsequent studies (Ostrom and Hoole, 1978; Levy, 1984; Siverson and Sullivan, 1984; Wayman, 1990; King, 1989; Kegley and Raymond, 1982) there are few robust finding to indicate whether alliance formation makes war more or less likely.

The link between alliance formation and the occurrence of war is ambiguous. However, alliances expand conflict. If a nation has an alliance with a combatant, then it is more likely to intervene in a conflict (Sabrosky, 1980; Siverson and King, 1980; Altfeld and Bueno de Mesquita, 1979; Kim, 1991; Oren, 1989; Siverson and Tennefoss, 1984; Smith, 1995a, b). Although alliances significantly increase the probability of intervention, many alliances are unreliable. Empirically, only about 25% of nations intervene on behalf of allies that become involved in wars (Siverson and King, 1980).

The low reliability of alliances is worrying in the light of theories of alliance formation. Many of these theories assume that alliances are reliable (Niou and Ordeshook, 1991, 1989; Altfeld, 1988; Conybeare, 1994; Morrow, 1986). There are many motivations for alliance formation. However, most alliances have some security motivations. It is these security concerns that I consider.

Rather than examine alliance formation, alliance reliability, and the effect of alliances as separate questions, I model these three aspects simultaneously. Nations use alliances to signal their intention to intervene should a war occur. Although not all alliances are reliable, upon seeing an alliance nations anticipate that intervention is likely. Alliance formation deters A and reassures B. This means that less attacks occur if an alliance forms. However, once an attack takes place, it is more likely that war occurs because B is less likely to surrender. Alliances are therefore related to the occurrence of war. When nations form alliances they do so knowing the reliability of the alliance, and the likely effect of alliance formation on the behavior of other states.

In the next section, I describe a formal model of crisis behavior and alliance formation. Examining the crisis model reveals how the anticipated behavior in one nation affects the behavior of other nations. I describe how a third party's decision to intervene in a conflict, a protege's decision to resist an attack and an aggressor's decision to attack are related. Thus, the model relates the probability of intervention with the probability of attack and the probability of resistance. Specifically, the model shows that the more likely the third party, C, is to intervene the more likely the protege is to resist. This increase in intervention and retaliation reduces the probability that an aggressor will attack. These results have important consequences for evaluating empirical evidence. I discuss these implications.

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Having shown that the probability of intervention affects the behavior of states in crisis situations, I consider the role of alliances in altering behavior. I consider the probability with which nations form alliances, and the reliability of those alliance that form. The model shows that alliance formation increases the probability of intervention. Thus, I discuss the effect of alliances in immediate extended deterrence.

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THE MODEL

Formally, there is a set of three nations: Nations $= \{A, B, C\}$. These nations are the attacker/aggressor, the protege/target and the defender/third party, respectively. The game form is shown in Fig. 1. The defender, C, decides whether to make a commitment to B.[1] This alliance commitment is non-binding, but costly. There are two possible alliance states: allied and non-allied, $l \in L = \{BC, \emptyset\}$, respectively. Having observed the alliance state, l, nation A decides whether to attack. If attacked the protege, nation B, decides whether to retaliate. If B retaliates then A and B are at war. In the event of a war, nation C must decide whether or not to intervene.

This is a game of incomplete information. I define a nation's type (Harsanyi, 1967–8) as the cost it incurs if it becomes involved in a war. For all nations, the cost of fighting is an element of the unit interval, $w_i \in [0, 1]$. Nations know the costs they face, but they only have beliefs about the costs that other nations face. For simplicity, I assume that these prior beliefs about cost are uniform over the unit interval. Thus nation i's

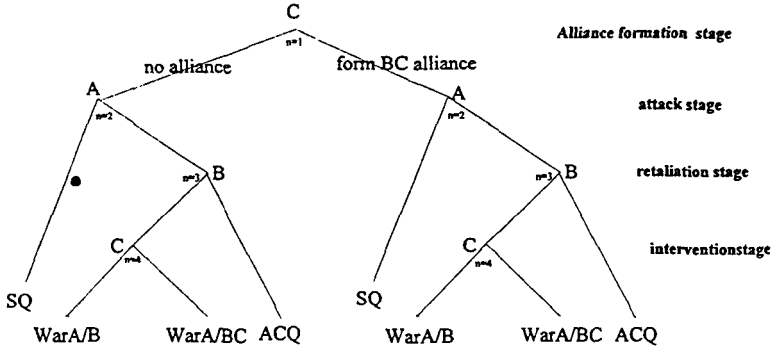


Figure 1. Alliance Formation with Incomplete Information

cost of fighting a war is $w_i \in [0,1]$. The other players do not know w_i, but their prior beliefs that the probability of $w_i \leqslant \tau$ (where $\tau \in [0,1]$) is τ.

Nations have preferences over outcomes. Table I shows the payoffs associated with each outcome. Each nation's payoff depends upon the outcome of the interaction and whether or not an alliance is formed.

Nations B and C are happy with the status quo and wish to retain it. However, nation A wants to move international policy from the status quo. In terms of how much they value the final policy, B and C's favorite outcome is SQ and their least preferred outcome is ACQ. Nation A has the opposite preferences; its most preferred outcome is ACQ, while its least preferred outcome is SQ.

How much a nation values a war depends upon who is likely to win and how costly it is to fight (a nation's type). The greater the probability that a nation wins, the greater its payoff from the war. If C intervenes to help nation B in its struggle against A then nation B is more likely to win. Therefore, nation B prefers a multilateral war to a bilateral war, while nation A prefers a bilateral war to a multilateral war: $y_b > x_b$ and $x_a > y_a$. Nation C has similar preferences to B; it wants to see nation B defeat nation A.

An important role of alliances is to enable nations to coordinate their military strategies (Morrow, 1994). Thus, nations that have an alliance might have a greater probability of winning a war. If B and C form an alliance before the war, then their expected value from a multilateral war might be higher than if they did not have an alliance. While B and C's ability to coordinate is to their advantage, it is to A's detriment: $z_b \geqslant y_b$, $z_c \geqslant y_c$ and $z_a \leqslant y_a$.

Thus in terms of the expected policy outcome alone, ignoring the costs of fighting and alliance formation, B and C's preferences from best to worst are SQ, WarA/BC with an alliance, WarA/BC with no alliance, WarA/C, ACQ: $1 > z_b \geqslant y_b > x_b > 0$, and $1 > z_c \geqslant y_c > x_c > 0$. Nation A has the opposite preference ordering over expected policy outcomes: $1 > x_a > y_a \geqslant z_a > 0$.

In addition to the expected policy outcome from an international crisis, nations care about the costs of fighting and the costs of alliances. As discussed above, the cost nation i pays for becoming involved in a war, w_i, is private information. Alliance politics is domestically and reputationally costly. Specifically, if C forms an alliance it pays a deadweight cost, f_c. Additionally, nation C pays a reputational and/or domestic cost if it does not support its ally, $h_c > 0$. Thus, there are both formation costs, f_c, and honor costs, h_c, associated with alliances. For the numerical example in Fig. 4 $y_b = y_c = z_b = z_c = 0.75$, $x_b = x_c = 0.4$, $x_a = 0.6$, $z_a = 0.25$ and $h_c = 0.3$.

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TABLE I
Payoffs for Nations

Nation	$U_i(\text{SQ}; 1)$	$U_i(\text{War}\,A/B; 1)$	$U_i(\text{War}\,A/BC; \emptyset)$	$U_i(\text{War}\,A/BC; BC)$	$U_i(\text{ACQ}; 1)$
A	0	$x_a - w_a$	$y_a - w_a$	$z_a - w_a$	1
B	1	$x_b - w_b$	$y_b - w_b$	$z_b - w_b$	0
C if $1 = \emptyset$	1	x_c	$y_c - w_c$	—	0
C if $1 = BC$	$1 - f_c$	$x_c - h_c - f_c$	—	$z_c - w_c - f_c$	$0 - f_c$

NOTE: $f_c > 0$, $h_c > 0$, $1 > x_a > y_a \geq z_a > 0$, $1 > z_b \geq y_b > x_b > 0$, and $1 > z_c \geq y_c > x_c > 0$.

AGGRESSION, RETALIATION AND INTERVENTION IN INTERNATIONAL CRISES

In this section I show how nation C's likely behavior influences other nations. Specifically, I show how the probability of intervention given alliance state l, $\gamma(l)$, affects the probability that A attacks, $\alpha(l)$, and the probability that B retaliates, $\beta(l)$. I start by showing that for any particular type of nation C the presence of an alliance makes it more likely to intervene. However, this is insufficient to show that alliance formation increases the probability of intervention because we do not yet know which types of C choose to form alliances. Rather than introduce the complexity of alliance formation at this stage, I simply characterize $\alpha(l)$ and $\beta(l)$ in terms of $\gamma(l)$. These relationships between $\alpha(l)$, $\beta(l)$, and $\gamma(l)$ are substantively important for analyzing empirical data. After exploring these implications, I characterize which nations form alliances and hence the reliability of alliances and their affect on the outcome of the crisis.

C's Decision to Intervene

First, consider the probability that nation C intervenes should a war occur. The expected utility C receives for intervening is

$$E[U_c(intervene\,|\,\emptyset)] = y_c - w_c, \quad \text{if non-allied } (l=\emptyset), \text{ and}$$

$$E[U_c(intervene\,|\,BC)] = z_c - w_c - f_c, \quad \text{if allied } (l=BC).$$

The expected utility of remaining neutral is

$$E[U_c(remain\ neutral\,|\,\emptyset)] = x_c, \quad \text{if non-allied } (l=\emptyset), \text{ and}$$

$$E[U_c(remain\ neutral\,|\,BC)] = x_c - h_c - f_c, \quad \text{if allied } (l=BC).$$

Nation C intervenes when the expected utility of doing so is greater than the expected utility of remaining neutral. Thus, if C is allied then it will intervene if $z_c - x_c + h_c \geq w_c$; if C is non-allied then it will intervene if $y_c - x_c + h_c \geq w_c$; otherwise C remains neutral. Perhaps not surprisingly, as the cost of fighting increases, C's incentive to intervene declines.

As C's cost of fighting increases, C is less likely to intervene. When the cost of fighting is low ($w_c \leq y_c - x_c$), C intervenes whether or not an alliance is formed. I call this set of types $t_1 = \{w_c\colon w_c \leq y_c - x_c\}$. However, as the cost of fighting increases ($z_c - x_c + h_c \geq w_c > y_c - x_c$), C needs the inducement of an alliance before it intervenes. An alliance encourages C to intervene in two ways. First, the military coordination aspects of

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alliances makes intervention more attractive because C expects to do better in the war. Second, the cost of failing to honor an alliance commitment, h_c, makes remaining neutral less attractive. Thus, the presence of an alliance can compel some types to intervene when they would not in the absence of an alliance. I call this set of types $t_2=\{w_c\colon z_c-x_c+h_c \geqslant w_c>y_c-x_c\}$.

As the cost of fighting becomes even higher ($w_c>z_c-x_c+h_c$), even an alliance is insufficient inducement for intervention. For these types, which I call $t_3=\{w_c\colon w_c>z_c-x_c+h_c\}$, the costs associated with war more than offset the honor cost of alliance failure.

Nations A and B do not know C's type. However, if they have beliefs about the distribution of C's type, then they can estimate the probability that nation C intervenes because they know how each type behaves. Alliance formation affects these beliefs. Later I show how the formation of an alliance determines the beliefs held by the other nations. The probability that nation C will intervene if an alliance forms, $\gamma(BC)$, is the probability that cost of fighting, w_c, is less than $z_c-x_c+h_c$. Without an alliance C intervenes only if $w_c<y_c-x_c$. Thus, the probability of non-allied intervention, $\gamma(\emptyset)$, is the probability that $w_c<y_c-x_c$ given that C choose not to form an alliance. I defer the discussion of how nations A and B calculate $\gamma(\emptyset)$ and $\gamma(BC)$ until later. What I want to show now is how $\gamma(\emptyset)$ and $\gamma(BC)$ affect the probability of A attacking and B retaliating.

B's Decision to Retaliate

If nation A attacks then nation B must decide whether or not to retaliate. If C did not offer an alliance commitment then B's expected utility from retaliating is

$$E[U_b(retaliate\,|\,s_c^*,\emptyset)]=\gamma(\emptyset)[y_b-w_b]+[1-\gamma(\emptyset)][x_b-w_b].$$

The first term, $\gamma(\emptyset)[y_b-w_b]$, is the probability that C intervenes, $\gamma(\emptyset)$, multiplied by B's payoff if C does intervene, (y_b-w_b). The second term, $[1-\gamma(\emptyset)][x_b-w_b]$, is the probability that C remains neutral multiplied by B's payoff for a bilateral war.

If nation B remains neutral then its payoff is

$$E[U_b(surrender\,|\,s_c^*,l)]=0.$$

Thus, when no alliance forms, nation B only retaliates if $w_b \leqslant \gamma(\emptyset)[y_b-x_b]+x_b$. Since B's cost of fighting, w_b, is distributed uniformly the probability that B retaliates is $\beta(\emptyset)=\gamma(\emptyset)[y_b-x_b]+x_b$.

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In the presence of an alliance, a similar calculation shows the B retaliates when $w_b \leqslant \gamma(BC)[z_b - x_b] + x_b$. Therefore, the probability that B retaliates is $\beta(BC) = \gamma(BC)[z_b - x_b] + x_b$.

A's Decision to Attack

Knowing the alliance state, nation A can calculate the probability of B retaliating, $\beta(l)$, and the probability of C intervening, $\gamma(l)$. Nation A uses the probabilities, β and γ, to decide whether to attack. If A attacks then with probability $\beta(l)\gamma(l)$ it becomes involved in a multilateral war against B and C, with probability $\beta(l)(1-\gamma(l))$ it becomes involved in a bilateral war against B, and with probability $(1-\beta(l))$ B surrenders.
Thus A's expected utility for attacking if no alliance forms is

$$E[U_a(attack \mid s_b^*, s_c^*, \emptyset)] = \beta(\emptyset)\gamma(\emptyset)(y_a - w_a) + \beta(\emptyset)(1-\gamma(\emptyset))(x_a - w_a) + (1-\beta(\emptyset)),$$

$$E[U_a(no\ attack \mid s_b^*, s_c^*, \emptyset)] = 0.$$

Rearranging these expressions reveals that A attacks if $w_a \leqslant \gamma(\emptyset)(y_a - x_a) + x_a - 1 + 1/\beta(\emptyset)$.

Given that A's type is distributed uniformly over the unit interval we can calculate the probability that nation A attacks:

$$\alpha(\emptyset) = \min\{\gamma(\emptyset)[y_a - x_a] + x_a - 1 + 1/\beta(\emptyset), 1\}.$$

The minimum function is simply to ensure that probabilities are restricted to the interval [0,1].

Similarly, if an alliance forms then A attacks if $w_a \leqslant \gamma(BC)(z_a - x_a) + x_a - 1 + 1/\beta(BC)$ and the probability of an attack is $\alpha(BC) = \min\{\gamma(BC) \times [z_a - x_a] + x_a - 1 + 1/\beta(BC), 1\}$.

THE RELATIONSHIP BETWEEN $\alpha(l)$, $\beta(l)$ AND $\gamma(l)$

PROPOSITION 1: For any alliance state, as the probability of intervention increases, then the probability of retaliation increases: $d\beta(l)/d\gamma(l) > 0$.

Proof: Follows directly from the definition of β.

PROPOSITION 2: For any alliance state, as the probability of intervention increases, then the probability of attack (weakly) decreases: $d\alpha(l)/d\gamma(l) \leqslant 0$.

Proof: Follows directly from Proposition 1 and the definition of α.

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Empirical Implications

The propositions above mean that as nation C becomes more likely to intervene nation A is less likely to attack and nation B is more likely to retaliate. These predictions, while not surprising, have profound empirical implications. In this section I explore these consequences. For brevity, I restrict discussion to the allied case, with the understanding that similar results hold in the non-allied case.

Nation A does not attack all nations with the same probability. The propositions above shows that nation A is most likely to attack irresolute nations with unreliable allies. Resolute nations (high β) with reliable allies (high γ) are not attractive targets for aggression. If A attacks these states then it is likely to find itself involved in a multilateral war. This is not to say that A never attacks resolute states with reliable allies; providing A anticipates the war going well (high z_a), it might be prepared to fight multilateral wars. Nation A attacks in those situations where it expects to be successful. As Siverson and Gartner (1996; see also Siverson, 1995; Wang and Ray, 1994; Smith, 1996) show initiators win most of the conflicts they start. The results of the model suggest that this is not because initiators have an inherent advantage. Rather initiators pick the fights in which they anticipate a successful conclusion.

A attacks when the probability of retaliation, β, and the probability of intervention, γ, are low. What are the empirical consequences of this? In order to answer this question systematically, consider a set of international events. Without becoming engaged in how to operationalize variables, suppose some events end in acquiesce, other lead to war and the remainder are status quo events. Suppose we are interested in explaining B and C's behavior. Unfortunately, we only observe B's behavior if A attacks. However, the model show that A's decision depends upon the probability that B retaliates. Therefore, the majority of events where A tests B's resolve, by attacking, are events where B is likely to back down. When alliances are reliable and B will resist, attacks are unlikely. Therefore, we do not observe the resolve of B or the reliability of C.

The average reliability of alliances, γ, and the resolve on targets, β, will be higher in status quo events than in events where attacks occur. We can only observe the reliability of alliances in cases where A attacks. However, the model predicts that the average reliability of alliances across all events is higher than the observed reliability. A's strategic behavior means that the sample of events in which we can observe alliance reliability- wars- contains mainly unreliable alliances.

Next consider B's decision to retaliate conditional upon being attacked. The model suggests that the more reliable an ally, the more likely B is to

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resist: As γ increases so does β. We can only observe whether C intervenes if a war actually occurs; but suppose we estimate alliance reliability conditional on A having attacked. Since B's decision to retaliate is related to the probability of intervention, the average alliance reliability in events that end in war is higher than the average reliability of acquiesce events. This prediction is supported by recent empirical tests (Smith, 1996). Alliance reliability is higher in cases of war than cases of acquiesce. This does not mean that B's decision affects the reliability of alliances, rather alliance reliability affects B's decision.

Unreliable alliance get attacked; reliable alliances don't. Unfortunately, these selection effects make statistical analysis difficult (Fearon, 1996). Suppose we want to examine alliance reliability, γ. We can only observe whether C intervenes if a war actually occurs; thus our sample is the set of wars. Unfortunately, the probability of intervention affects A and B's decisions. Thus, the value of the dependent variable, γ (the probability of intervention) affects whether an event is selected into the sample, the sets of war. The consequence of selecting on the dependent variable is that the standard econometric assumption that the expected value of the stochastic error term is zero is violated (Achen, 1986; Dubin and Rivers, 1989; Greene, 1990; Heckman, 1976, 1979).[2] Therefore, estimates of alliance reliability obtained using standard statistical procedure (such as probit) are biased.

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ALLIANCE FORMATION

The probability of intervention affects the behavior of nations A and B. If nation C can convince A and B that it will intervene, then the other actors change their behavior. Specifically, A attacks less and B retaliates more. In this section, I discuss when nations form alliances, and how immediate alliance formation affects the outcome of crises.

Nation C wants to maintain the status quo. Therefore, it wants to deter A from attacking and encourage B to resist any challenge by A. As I demonstrate above, C can fulfill these aims if it convinces A and B that it will intervene. Nation C uses alliance formation to increase the probability of intervention, γ. Alliance formation affects intervention, γ, through two mechanisms: commitment and signaling. I discuss each in turn.

Commitment

Having formed an alliance, intervention becomes more attractive relative to remaining neutral. Alliance formation affects C's payoffs in two ways.

First, the honor cost, h_c, of deserting an ally reduces C's payoff from non-intervention. Second, the coordination benefits of alliances ($z_c \geq y_c$) mean that, if allied, B and C are more likely to defeat A. This makes intervention more attractive. These two factors combine to increase the incentives for C to intervene if it is allied. By forming an alliance, C can commit itself to intervene. Having formed an alliance nation C might intervene whereas without an alliance it would not. Thus, conditional on the presence of an alliance, every type is more likely to intervene to avoid the costs of alliance failure.

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Signaling

Different types have different incentive to form alliances. Therefore, some types choose to form alliances while other types do not. If A and B observe an alliance form then they know that C is one of the types that forms alliances. In order to illustrate this consider the following situation. Suppose there is a type $w_c = \theta$, such that all types lower than θ form alliance and all types with cost of fighting greater than θ do not form an alliance. Upon seeing alliance formation, A and B know that C's cost of fighting is less than θ: $w_c \leq \theta$. If no alliance forms then they know that C's cost of fighting is higher than θ: $w_c > \theta$. Suppose type θ is a type that intervenes if it has an alliance but does not intervene otherwise: $\theta \in t_2$, $y_c - x_c < \theta < z_c - x_c + h_c$.

Since alliance formation indicates that $w_c \leq \theta$, A and B know that C will intervene having formed an alliance. Therefore, the probability of intervention given that an alliance forms is one: $\gamma(BC) = 1$. If no alliance forms then A and B know that the probability of intervention is zero, since the types that choose not to form an alliance face high cost for fighting and hence will not intervene: $\gamma(\emptyset) = 0$.

Alliance formation affects the probability of intervention. It allows nations to commit. It also allows nations to identify their type. The example above illustrates how this updating about type works. Next I present the main result about the affect of alliance formation on the probability of intervention. Using this result I show the incentives to form an alliance, and discuss the equilibrium solutions.

Alliance Formation and the Probability of Intervention

PROPOSITION 3: Alliance formation increases the probability of intervention: $\gamma(BC) > \gamma(\emptyset)$.

Proof: See the Appendix.

Before considering the substantive implications of Proposition 3, I want to illustrate what drives the results and characterize alliance formation equilibria. Consider nation C's payoff if it forms an alliance. With probability $\alpha(BC)\beta(BC)$ a war occurs. When war occur C's payoff depends upon whether it intervenes. If $w_c \in t_3 (w_c > z_c - x_c + h_c)$ then C remains neutral and pays the alliance failure cost h_c. Thus, its payoff is $x_c - h_c$. If C's cost of fighting is lower, $w_c \in t_1 \cup t_2$, then C intervenes and its payoff is $z_c - w_c$. With probability $\alpha(BC)(1-\beta(BC))$, A attacks and B surrender. C's payoff for acquiesce is 0. With probability $(1-\alpha(BC))$, A does not attack. C's payoff for the status quo is 1. Finally, C pays an alliance formation cost, f_c. Therefore, C expected payoff from alliance formation is

$$E[U_c(BC; w_c)] = \alpha(BC)\beta(BC)V_c(BC, w_c) + \alpha(BC)(1-\beta(BC))0 + (1-\alpha(BC)) - f_c,$$

where $V_c(BC, w_c) = z_c - w_c$ if $w_c \in t_1 \cup t_2$ and $V_c(BC, w_c) = x_c - h_c$ if $w_c \in t_3$.

If C does not form an alliance then war occurs with probability $\alpha(\emptyset)\beta(\emptyset)$. C's payoff for war depends upon its type. Those types with low cost of fighting ($w_c \in t_1$; $w_c \leqslant y_c - x_c$) have a payoff of $y_c - w_c$; higher cost types receive a payoff of x_c. C's payoff for acquiesce and status quo are zero and one, respectively. Therefore C's expected payoff if no alliance forms is

$$E[U_c(\emptyset; w_c)] = \alpha(\emptyset)\beta(\emptyset)V_c(\emptyset, w_c) + \alpha(\emptyset)(1-\beta(\emptyset))0 + (1-\alpha(\emptyset)),$$

where $V_c(\emptyset, w_c) = y_c - w_c$ if $w_c \in t_1$ and $V_c(\emptyset, w_c) = x_c$ if $w_c \in t_2 \cup t_3$.

Characterizing Alliance Formation

Let $\sigma_c(w_c)$ represent an alliance formation strategy, where $\sigma_c(w_c)$ is the probability that type w_c forms an alliance. Given any alliance formation strategy, A and B can calculate their posterior beliefs about the type of C using Bayes rule. Further, since A and B know whether each type will intervene should a war occur, they can calculate the probability of intervention: $\gamma(BC)$ and $\gamma(\emptyset)$. In the example above with $\theta \in t_2$, an alliance implies that $w_c < \theta$. Since all types less that $z_c - x_c + h_c$ will intervene given an alliance, $\gamma(BC) = 1$. Similar calculations can be performed for any alliance formation strategy. The general calculation is presented in the appendix.

For any alliance formation strategy, $\sigma_c(w_c)$, C can calculate the value of forming an alliance versus the value of not forming an alliance. Define

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Diff(σ_c, w_c) as the expected utility difference between forming and not forming an alliance: Diff(σ_c, w_c) = $E[U_c(BC; w_c)] - E[U_c(\emptyset; w_c)]$. Thus, if Diff($\sigma_c, w_c$) is positive then (given the beliefs created by the alliance formation strategy σ_c) type w_c prefers to form an alliance rather than remain non-allied.

If $\sigma_c(w_c)^*$ is an equilibrium alliance formation strategy then if type w_c forms an alliance, $\sigma_c(w_c)^* > 0$, then Diff(σ_c^*, w_c) ≥ 0. Characterizing the alliance formation equilibria involves finding those strategies, σ_c, where only those types of C that want to form alliances actually do. This search is simplified by looking at the shape of Diff(σ_c, w_c). Differentiating Diff(σ_c, w_c) with respect to type reveals that

$$\frac{\partial \text{Diff}(\sigma_c, w_c)}{\partial w_c} = -\alpha(BC)\beta(BC) + \alpha(\emptyset)\beta(\emptyset) \quad \text{if } w_c \in t_1,$$

$$= -\alpha(BC)\beta(BC) < 0 \quad \text{if } w_c \in t_2, \text{ and}$$

$$= 0 \quad \text{if } w_c \in t_3.$$

This is shown pictorially in Fig. 2. For types with low costs of fighting, $w_c < y_c - x_c$, Diff(σ_c, w_c) can be either increasing or decreasing with respect to type. For types $w_c \in t_2$, Diff(σ_c, w_c) is strictly decreasing. Finally, for those types with high cost of fighting, $w_c \in t_3$, Diff(σ_c, w_c) is flat.

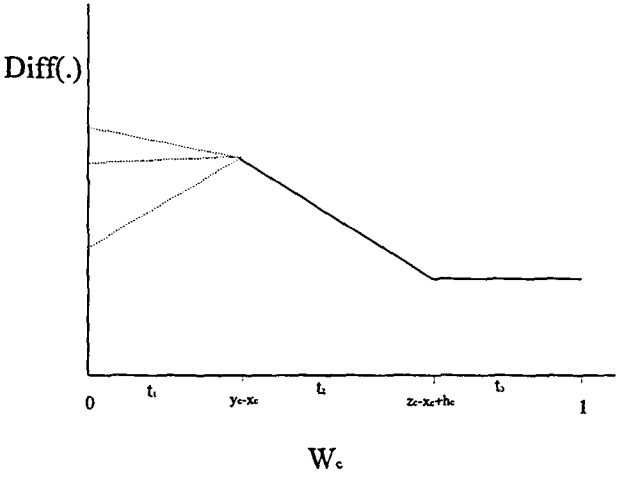


Figure 2. The Shape of Diff(·)

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Suppose that alliance formation makes war more likely: $\alpha(BC)\beta(BC) \geqslant \alpha(\emptyset)\beta(\emptyset)$. Diff$(\sigma_c, w_c)$ has negative slope. Under these conditions, if any type θ is indifferent about alliance formation, then all types less that θ form an alliance and all types greater than θ do not form an alliance.[3] This characterization describes five of the seven types of equilibria that can occur in this game. I label these equilibria E0, E1, E2, E3 and E4. The Appendix contains a full characterization of all the equilibria. Figure 3 is a pictorial representation of some of these equilibria. Their properties are summarized in Figure 5.

In equilibrium E0, $\theta = 0$. No types form alliances. Even though alliances could have a substantial effect, deterring A, the cost of alliance formation

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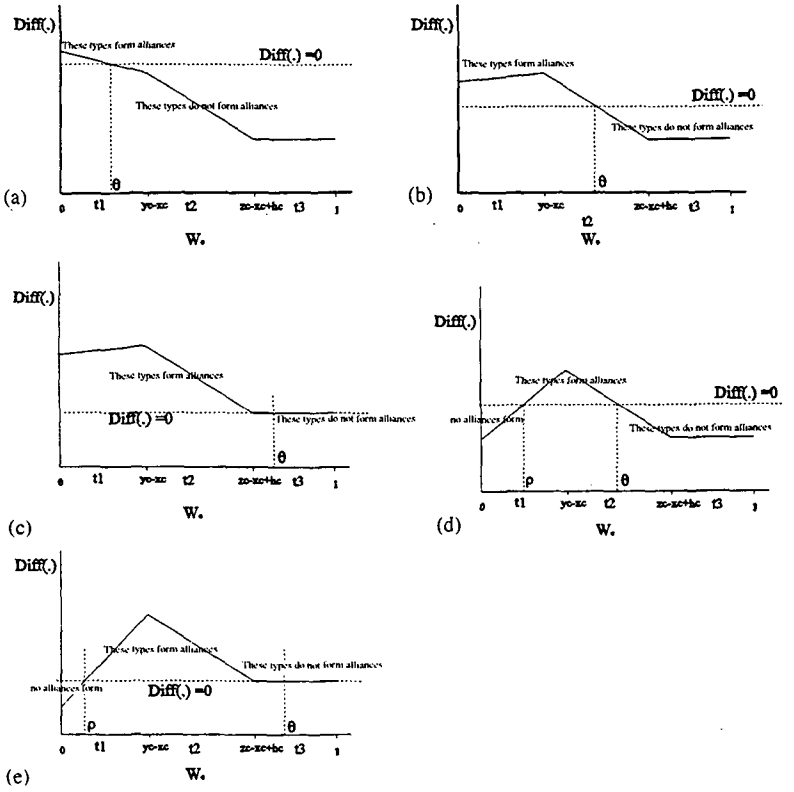


Figure 3. (a) Equilibrium E1, (b) Equilibrium E2, (c) Equilibrium E3, (d) Equilibrium E2b, (e) Equilibrium E3b

outweighs the benefits. In the numerical example shown in Fig. 4 this equilibrium occurs as the cost of alliance formation rises above $f_c = 0.383$.

In equilibrium E1, $\theta \in t_1$. Only a small proportion of nations form alliances but all these alliances are reliable. If no alliance forms then some nations will still intervene: Specifically, $\gamma(\emptyset) = (y_c - x_c - \theta)/(1-\theta)$. In equilibrium E1 alliance formation makes war more likely.

Equilibrium E2 is the case I used to illustrate the signaling of alliance formation above. $\theta \in t_2$. All types that form alliances are prepared to intervene. For the types $w_c > y_c - x_c$, it is the commitment of alliance failure costs that makes them honor their alliance. Non-allied intervention never occurs in this equilibrium. Thus, $\gamma(BC) = 1$ and $\gamma(\emptyset) = 0$.

In equilibrium E3 the cost of alliance formation is sufficiently low that some types will form alliances to alter A and B's behavior even though they have no intention of intervening. Thus, $\theta \in t_3$. For those types $w_c > z_c - x_c + h_c$ that form alliances the cost of fighting is so high that it offsets the cost of alliance failure. In equilibrium E3 some alliances are unreliable but the failure to form an alliance is a definite signal of non-intervention: $\gamma(BC) = (z_c - x_c + h_c)/\theta$ and $\gamma(\emptyset) = 0$.

If the cost of alliance formation is extremely low then all types form an alliance, $\theta = 1$. In this equilibrium, which I call E4, alliance reliability is $\gamma(BC) = z_c - x_c + h_c$. Since every type forms an alliance, there is no signaling

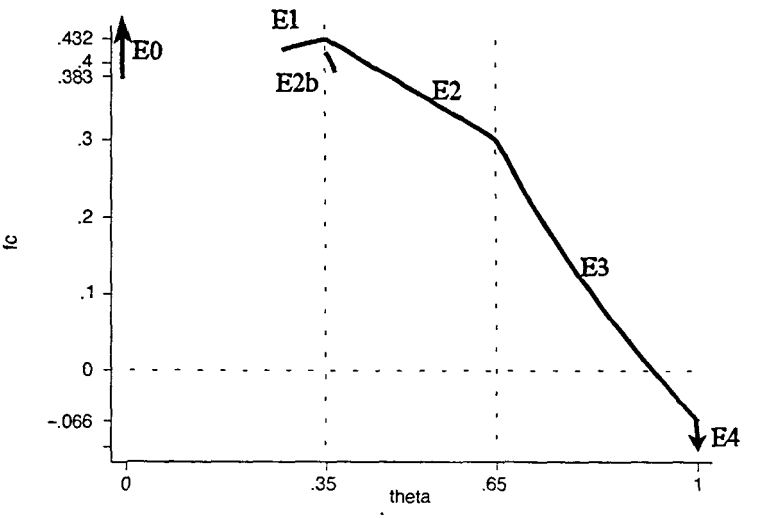


Figure 4. Numerical Example of Alliance Formation Equilibria

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Equilibrium	Proportion of nation C's forming alliance		Alliance reliability, $\gamma(BC)$		Probability of intervention without alliance, $\gamma(\emptyset)$		Probability of attack if an alliance forms, $\alpha(BC)$	Probability of retaliation if an alliance forms, $\beta(BC)$
E0	0	least	1	most	$y_c - x_c$	most	least	most
E1	$0 < \theta \leqslant y_c - x_c$	↓	1	↑	$(y_c - x_c - \theta)/(1-\theta)$	↑	↓	↑
E2	$y_c - x_c < \theta \leqslant z_c - x_c + h_c$		1		0			
E3	$z_c - x_c + h_c < \theta < 1$		$(z_c - x_c + h_c)/\theta$		0			
E4	1	most	$z_c - x_c + h_c$	least	0	least	most	least

Figure 5. Properties of some of the Equilibria

advantage to forming an alliance. Nations cannot use an alliance as a signal that they have a low cost of fighting because every type uses this message. In this equilibrium, alliance formation functions only as a commitment mechanism.

If alliance formation reduces the probability of war then there are two other equilibria: E2b and E3b. For nation C the advantage of forming an alliance is that it deters A and reassures B. Loosely speaking, if A is more deterrable than B is reassurable then alliance formation reduces the incidence of war. If C pays the alliance formation costs then its gains from deterring A. However, for those types with the very lowest cost of fighting, w_c close to zero, the cost of alliance formation does not offset the advantages of deterring A. Facing very low costs of fighting, these types are happy to see wars occur rather than pay alliance formation costs to prevent them. Equilibria E2b and E3b are shown in Fig. 3. Their formal characterization is in the Appendix. No examples of E3b occur in the numerical example in Fig. 4 but examples of E2b are shown.

The Implications of Alliance Formation

Alliance formation increases the probability of intervention: $\gamma(BC) > \gamma(\emptyset)$. An allied third party is more likely to enter a war than a non-allied third party. Indeed, this finding fits the empirical evidence well (Altfeld and Bueno de Mesquita, 1979). The probability of intervention affects A and B's decisions. Therefore, alliance formation affects both the probability that nation B retaliates and the probability that nation A attacks. Specifically, alliances deter aggressors and reassure targets.

PROPOSITION 4: Alliance formation increases the probability that B resists an attack: $\beta(BC) > \beta(\emptyset)$.

Proof: $\beta(BC) = \gamma(BC)(z_b - x_b) + x_b$ and $\beta(\emptyset) = \gamma(\emptyset)(y_b - x_b) + x_b$. By Proposition 3, $\gamma(BC) > \gamma(\emptyset)$. This implies that $\beta(BC) > \beta(\emptyset)$.

PROPOSITION 5: Alliance formation weakly reduces the probability that nation A attacks: $\alpha(BC) \leqslant \alpha(\emptyset)$.

Proof: $\alpha(BC) = \min\{1, \gamma(BC)(z_a - x_a) + x_a - 1 + 1/\beta(BC)\}$ and $\alpha(\emptyset) = \min\{1, \gamma(\emptyset)(y_a - x_a) + x_a - 1 + 1/\beta(\emptyset)\}$. By Propositions 3 and 4 $\gamma(BC) > \gamma(\emptyset)$ and $\beta(BC) > \beta(\emptyset)$. Therefore, $\gamma(BC)(z_a - x_a) + x_a - 1 + 1/\beta(BC) < \gamma(\emptyset)(y_a - x_a) + x_a - 1 + 1/\beta(\emptyset)$. This implies that $\alpha(BC) \leqslant \alpha(\emptyset)$.

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EMPIRICAL IMPLICATIONS

Alliances influence the occurrence of conflict. The formation of an alliance makes nation A less likely to attack and nation B more likely to resist. Empirically, we expect non-allied nations to be attacked at a higher rate than allied nations. Siverson and Tennefoss (1984, Table V, p. 1063) find empirical support for this claim. Examining major versus minor power conflict, they find that major power initiators attacked 87 non-allied minor powers but attacked only 39 allied minor powers. The model predicts that there should also be a large number of cases where non-allied minor powers concede to the demands of major power initiators.

Gartner and Siverson (1996; Siverson, 1995) find that the intervention of third parties has a large effect on the outcome of conflicts. Specifically, they find that targets that receive third party help won 8 out of 15 conflicts. However, targets receiving no aid won only 9 out of 44 conflicts. Non-allied states receive third party support at lower rates than allied states. In 10 of 17 conflict events allied states receive support. This compares with only 5 of 42 non-allied states receiving support (Gartner and Siverson, 1996, Table II). Allied states receive more intervention. They are also attacked less often as the model predicts (27 allied nations versus 47 non-allied nations are attacked).

Alliances also affect the distribution of outcomes. If an alliance forms, then A attacks less and B retaliates more. Thus, crises preceded by alliance formation are likely to result in war or status quo. Compared to crises where no alliances form, they are less likely to end in acquiesce. Without an alliance, crises are more likely to end in war and acquiesce, but the status quo is less likely.

The Alliance/War Relationship

The formation of an alliance has predictable effects on the behavior of the potential defender, the target and the aggressor. However, despite this, there is no general relationship between the formation of an alliance and the occurrence of war. The probability of war is $\alpha\beta$. Alliance formation reduces the probability of A attacking. This reduces the likelihood of war. However, alliance formation also increases the probability that B will retaliate if attacked. This increases the likelihood of war. In some situations alliance formation may increase the probability of war. Yet, in other situations alliance formation may reduce the probability of armed conflict.

In general, it is ambiguous whether $\alpha(BC)\beta(BC)$ or $\alpha(\emptyset)\beta(\emptyset)$ is larger. In equilibrium E1 the probability of war increases with alliance formation.

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In equilibria E2b and E3b the probability of war decreases with alliance formation. In the other equilibria either case is possible. In order to determine whether alliance formation increases or decreases the probability of war, it is necessary to examine the conditions to see which equilibria are likely to exist. If alliance formation has a larger predicted effect on nation *A* than nation *B*, then alliance formation is likely to result in a lower probability of war, and vice versa. This result suggests that empirical tests looking at the correlation between alliance formation and the occurrence of war are unlikely to find an aggregate effect. On the whole these empirical tests have been inconclusive (Singer and Bueno de Mesquita, 1973; Ostrom and Hoole, 1978; Wayman, 1990; Levy, 1984; King, 1989). Although the aggregate effect of alliances on the occurrence of war is ambiguous, the effect of alliances on aggressor and target nations is predictable.

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CONCLUSIONS

Alliance formation reassures friends, making them more likely to retaliate. Alliance formation also deters enemies, making them less likely to attack. Alliance formation affects the behavior of aggressors and targets because it alters the expected behavior of third parties. Specifically, an alliance increases the probability of intervention should a war occur.

This paper relates alliance formation to a model of war in an incomplete information setting. From this model, I develop a theory to explain the effects of alliances. The theory predicts how alliances affect the behavior of nations, and when nations form alliances. Deductive modeling is important in interpreting empirical evidence. This paper provides an interpretation of existing empirical evidence. It also provides new testable hypotheses. Modeling specifies the relationships to be tested. This allows the application of suitable econometric techniques, and suggests when traditional tests are inappropriate.

APPENDIX

This appendix formally characterizes the alliance formation equilibria. Before discussing these equilibria, I clarify several concepts that are used in the characterization.

The solution concept is sequential equilibrium (Kreps and Wilson, 1982). In order to characterize the pooling equilibria I apply restrictions on

the out of equilibrium beliefs. These refinements are discussed later in the Appendix.

The Probability of Intervention

Nations A and B can calculate the probability of intervention depending upon whether or not an alliance forms. In the absence of an alliance, C intervenes if $w_c < y_c - x_c$; otherwise C remains neutral. In the presence of an alliance, C intervenes if $w_c < z_c - x_c + h_c$.

In general, given any alliance formation strategy, $\sigma_c(w_c)$, the probability that C intervenes should a war occur can be calculated using Bayes rule. The probability that C intervenes if it does not form an alliance is $\gamma(\emptyset)$.

$$\gamma(\emptyset) = \text{Prob} \cdot [w_c < y_c - x_c | \emptyset] = \frac{\text{Prob}[\emptyset | w_c < y_c - x_c] \cdot \text{Prob}[w_c]}{\text{Prob[no alliance]}},$$

$$\gamma(\emptyset) = \int_0^{y_c - x_c} (1 - \sigma_c(w_c))\, dw_c \Big/ \int_0^1 (1 - \sigma_c(w_c))\, dw_c.$$

The probability that C intervenes if it forms an alliance can be calculated in a similar manner.

$$\gamma(BC) = \int_0^{z_c - x_c + h_c} \sigma_c(w_c)\, dw_c \Big/ \int_0^1 \sigma_c(w_c)\, dw_c.$$

For any alliance formation strategy, $\sigma_c(w_c)$, the beliefs $\gamma(\emptyset)$ and $\gamma(BC)$ can be calculated as described above. Given an alliance formation strategy, $\sigma_c(w_c)$, we can calculate the expected utility that each type obtains from forming an alliance.

The expected utility C receives from forming an alliance is

$$E[U_c(BC; w_c)] = \alpha(BC)[\beta(BC)\, V_c(BC, w_c) - 1] + 1 - f_c,$$

where $V_c(BC, w_c) = z_c - w_c$ if $w_c \in t_1 \cup t_2$ and $V_c(BC, w_c) = x_c - h_c$ if $w_c \in t_3$.

If C does not form an alliance then its expected payoff is

$$E[U_c(\emptyset; w_c)] = \alpha(\emptyset)[\beta(\emptyset) V_c(\emptyset, w_c) - 1] + 1,$$

where $V_c(\emptyset, w_c) = y_c - w_c$ if $w_c \in t_1$ and $V_c(\emptyset, w_c) = x_c$ if $w_c \in t_2 \cup t_3$. $V_c(1, w_c)$ is C's payoff if a war occurs.

Define $\text{Diff}(\sigma_c, w_c)$ as the expected utility difference between forming and not forming an alliance for type w_c, given the alliance formation strategy σ_c:

$$\text{Diff}(\sigma_c, w_c) = E[U_c(BC; w_c)] - E[U_c(\emptyset; w_c)].$$

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LEMMA 1: Given the definitions of $\alpha(l)$, $\beta(l)$ and $\gamma(l)$, $\sigma_c(w_c)^*$ is a sequential equilibrium iff $s_c(w_c)^* > 0$ implies $\text{Diff}(\sigma_c, w_c) \geq 0$ and $\sigma_c(w_c)^* < 1$ implies $\text{Diff}(\sigma_c, w_c) \leq 0$.

Proof: Sequential equilibrium requires that every player's strategy is utility maximizing given the strategy of the other players and their beliefs. $\alpha(l)$, $\beta(l)$ and $\gamma(l)$ are defined such that beliefs are consistent with Bayes rule and that A's decision to attack, B's decision to retaliate and C's decision to intervene are utility maximizing. Therefore all that remains is to show that C's decision to ally is utility maximizing given $\alpha(l)$ and $\beta(l)$.

Suppose $\sigma_c(w_c)^* > 0$ but $\text{Diff}(\sigma_c, w_c) < 0$. This implies that C's payoff from forming an alliance is lower than not forming an alliance. This contradicts $\sigma_c(w_c)^*$ being utility maximizing. Now suppose $\sigma_c(w_c)^* < 1$ but $\text{Diff}(\sigma_c, w_c) > 0$. This implies that C's payoff from not forming an alliance is lower than forming an alliance. This contradicts $\sigma_c(w_c)^*$ being utility maximizing. Next, if C's alliance formation strategy is utility maximizing then $\sigma_c(w_c)^*$ is a sequential equilibrium because every player's strategy is utility maximizing and beliefs are consistent with Bayes rule.

We can use the $\text{Diff}(\cdot)$ function to examine the incentives for different types to form alliances. Differentiating $\text{Diff}(\sigma_c(w_c), w_c)$ with respect to w_c reveals that Diff is either a weakly decreasing monotone or is single peaked with a maximum at $w_c = y_c - x_c$. This is pictorially demonstrated in Fig. 2.

$$\frac{\partial \text{Diff}(\sigma_c, w_c)}{\partial w_c} = -\alpha(BC)\beta(BC) + \alpha(\emptyset)\beta(\emptyset) \quad \text{if } w_c \in t_1,$$

$$= -\alpha(BC)\beta(BC) < 0 \quad \text{if } w_c \in t_2, \text{ and}$$

$$= 0 \quad \text{if } w_c \in t_3.$$

Alliance Formation Equilibria

I characterize alliance formation equilibria using two parameters: ρ and θ. Those types between ρ and θ form alliances, while those types greater than θ or less than ρ do not form alliances: If $w_c \in [p, \theta]$ then $\sigma_c(w_c) = 1$; if $w_c > \theta$ then $\sigma_c(w_c) = 0$; if $w_c < \rho$ then $\sigma_c(w_c) = 0$. There are seven types of equilibria in this game. They are summarized in the Table II below.

Equilibrium E0 is a pooling equilibrium in which no alliance forms. Equilibrium E4 is a pooling equilibrium in which all types form alliances. The remaining equilibria are semi-separating with some types forming alliances and others not. The semi-separating equilibria are shown in Fig. 3. Given the shape of $\text{Diff}(\sigma_c, w_c)$ these are the only possible equilibria.

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TABLE II
Alliance Formation Equilibria

Equilibrium	ρ	θ	$\gamma(\emptyset)$	$\gamma(BC)$
E0	0	0	$y_c - x_c$	1
E1	0	$\theta \in t_1$	$(y_c - x_c - \theta)/(1-\theta)$	1
E2	0	$\theta \in t_2$	0	1
E3	0	$\theta \in t_3$	0	$(z_c - x_c + h_c)/\theta$
E4	0	1	0	$z_c - x_c + h_c$
E2b	$\rho \in t_1$	$\theta \in t_2$	$\rho/(1+\rho-\theta)$	1
E3b	$\rho \in t_1$	$\theta \in t_3$	$\rho(1+\rho-\theta)$	$(z_c - x_c + h_c - \rho)/(\theta-\rho)$

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Semi-Separating Equilibria

E1: $\sigma_c^*(w_c < \theta) = 1$ and $\sigma_c^*(w_c > \theta) = 0$ where $\theta \in t_1$ such that $\text{Diff}(\sigma_c^*, \theta) = 0$ and $\alpha(BC)\beta(BC) + \alpha(\emptyset)\beta(\emptyset) \leqslant 0$.

Proof: $-\alpha(BC)\beta(BC) + \alpha(\emptyset)\beta(\emptyset) \leqslant 0$ implies that $\text{Diff}(\sigma_c, w_c)$ is decreasing with respect to w_c. Thus, if type θ is indifferent between forming an alliance or not, all types greater than θ prefer to remain non-allied. For $w_c < \theta$ alliance formation is utility maximizing.

E2: $\sigma_c^*(w_c < \theta) = 1$ and $\sigma_c^*(w_c > \theta) = 0$, where $\theta \in t_2$ such that

$$\text{Diff}(\sigma_c^*, \theta) = 0 \text{ and}$$

$$\text{Diff}(\sigma_c^*, w_c = 0) = \alpha(BC)[\beta(BC)z_c - 1] - \alpha(\emptyset)[\beta(\emptyset)z_c - 1] - f_c \geqslant 0.$$

Proof: Given that type $\theta \in t_2$ is indifferent about alliance formation and $\text{Diff}(\sigma_c, w_c)$ is decreasing over the interval $t_2 \cup t_3$, all types greater than θ prefer to remain non-allied. Next show that all $w_c < \theta$ want to form alliances. If $\alpha(BC)\beta(BC) \geqslant \alpha(\emptyset)\beta(\emptyset)$ then $\text{Diff}(\sigma_c, w_c)$ is decreasing over its whole range. Thus, $\text{Diff}(\sigma_c, w_c) > 0$ for all $w_c < \theta$. If $\alpha(BC)\beta(BC) < \alpha(\emptyset)\beta(\emptyset)$ then $\text{Diff}(\sigma_c, w_c)$ is increasing over t_1. Therefore, the type in t_1 with the least incentive to ally is $w_c = 0$. But $\alpha(BC)[\beta(BC)z_c - 1] - \alpha(\emptyset)[\beta(\emptyset)z_c - 1] - f_c \geqslant 0$ ensures that this type want to ally. Thus, $\sigma_c^*(w_c) = 1$ for all $w_c < \theta$.

E3: $\sigma_c^*(w_c < \theta) = 1$ and $\sigma_c^*(w_c > \theta) = 0$, where $\theta \in t_2$ such that

$$\text{Diff}(\sigma_c^*, \theta) = 0 \text{ and}$$

$$\text{Diff}(\sigma_c^*, w_c = 0) = \alpha(BC)[\beta(BC)z_c - 1] - \alpha(\circledcirc)[\beta(\circledcirc)z_c - 1] - f_c \geqslant 0.^3$$

Proof: Given that $\theta \in t_3$ is indifferent about alliance formation and $\text{Diff}(\sigma_c, w_c)$ is flat over the interval t_3, $\text{Diff}(\sigma_c, w_c) = 0$ for all $w_c \in t_3$. Next show that all $w_c < \theta$ $\text{Diff}(\sigma_c, w_c) \geqslant 0$. Since $\text{Diff}(\sigma_c, w_c)$ is decreasing or single peaked at $y_c - x_c$, if $\text{Diff}(\sigma_c^*, w_c = 0) = \alpha(BC)[\beta(BC)z_c - 1] - \alpha(\circledcirc)[\beta(\circledcirc)z_c - 1] - f_c \geqslant 0$ then for all $w_c \in [0, \theta]$ $\text{Diff}(\sigma_c, w_c) \geqslant 0$.

E2b: $\sigma_c^*(\rho \leqslant w_c \leqslant \theta) = BC$, $\sigma_c^*(w_c < \rho) = \emptyset$ and $\sigma_c^*(w_c > \theta) = \emptyset$, where $\rho \in t_1$, $\theta \in t_2$, such tha t

$\text{Diff}(\sigma_c^*, \rho) = \alpha(BC)[\beta(BC)[z_c - \rho] - 1] - \alpha(\emptyset)[\beta(\emptyset)(y_c - \rho) - 1] - f_c = 0$ and

$\text{Diff}(\sigma_c^*, \theta) = \alpha(BC)[\beta(BC)[z_c - \theta] - 1] - \alpha(\emptyset)[\beta(\emptyset)x_c - 1] - f_c = 0$.

Proof: If $\text{Diff}(\sigma_c^*, \rho) = \text{Diff}(\sigma_c^*, \theta) = 0$ then $\text{Diff}(\sigma_c^*, w_c)$ is single peaked at $y_c - x_c$. This implies that for all $w_c \in [\rho, \theta]$ $\text{Diff}(\sigma_c^*, w_c) \geqslant 0$, for $w_c < \rho$ $\text{Diff}(\sigma_c^*, w_c) < 0$, and for $w_c > \theta$, $\text{Diff}(\sigma_c^*, w_c) < 0$.

E3b: $\sigma_c^*(\rho \leqslant w_c \leqslant \theta) = 1$, $\sigma_c^*(w_c < \rho) = 0$ and $\sigma_c^*(w_c > \theta) = 0$ where $\rho \in t_1$ and $\theta \in t_3$ such that

$\text{Diff}(\sigma_c^*, \rho) = \alpha(BC)[\beta(BC)(z_c - \rho) - 1] - \alpha(\emptyset)[\beta(\emptyset)(y_c - \rho) - 1] - f_c = 0$ and

$\text{Diff}(\sigma_c^*, \theta) = \alpha(BC)[\beta(BC)[x_c - h_c] - 1] - \alpha(\emptyset)[\beta(\emptyset)x_c - 1] - f_c = 0$.

Proof: If $\text{Diff}(\sigma_c^*, \rho) = \text{Diff}(\sigma_c^*, \theta) = 0$ then the function $\text{Diff}(\sigma_c^*, w_c)$ is single peaked at $y_c - x_c$. This implies that for all $w_c \in [\rho, \theta]$, $\text{Diff}(\sigma_c^*, w_c) \geqslant 0$, for $w_c < \rho$, $\text{Diff}(\sigma_c^*, w_c) < 0$, and for $w_c > \theta$, $\text{Diff}(\sigma_c^*, w_c) = 0$.

Pooling Equilibria

E0: $\sigma_c^*(w_c) = 0$ for all w_c, if $\text{Diff}(\sigma_c^*, w_c = 0) = [\alpha(BC)[\beta(BC)z_c - 1] - \alpha(\emptyset)[\beta(\emptyset)y_c - 1]] - f_c \leqslant 0$ and $\text{Diff}(\sigma_c^*, w_c = y_c - x_c) = [\alpha(BC)[\beta(BC)z_c - 1] - \alpha(\emptyset)[\beta(\emptyset)x_c - 1]] - f_c \leqslant 0$. In the numerical example, I assume the out of equilibrium belief $\gamma(BC) = 1$.

Proof: $\text{Diff}(\sigma_c^*, w_c)$ reaches its maximum at either $w_c = y_c - x_c$ or $w_c = 0$. Thus, if $\text{Diff}(\sigma_c^*, w_c = y_c - x_c) \leqslant 0$ and $\text{Diff}(\sigma_c^*, w_c = 0) \leqslant 0$ then $\text{Diff}(\sigma_c^*, w_c) \leqslant 0$ for all w_c.

E4: $\sigma_c^*(w_c) = 1$ for all w_c, if $\text{Diff}(\sigma_c^*, w_c = 0) = [\alpha(BC)[\beta(BC)z_c - 1] - \alpha(\emptyset)[\beta(\emptyset)y_c - 1]] - f_c \geqslant 0$ and $\text{Diff}(\sigma_c^*, w_c \in t_3) = [\alpha(BC)[\beta(BC)x_c - 1] - \alpha(\circledcirc)[\beta(\circledcirc)x_c - 1]] - f_c \geqslant 0$. In the numerical example, I assume the out of equilibrium belief $\gamma(\emptyset) = 0$.

Proof: $\text{Diff}(\sigma_c^*, w_c)$ reaches its minimum at either $w_c = 0$ or $w_c \in t_3$. Thus, if $\text{Diff}(\sigma_c^*, w_c = 0) \geqslant 0$ and $\text{Diff}(\sigma_c^*, w_c \in t_3) \geqslant 0$ then $\text{Diff}(\sigma_c^*, w_c) \geqslant 0$ for all w_c.

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PROPOSITION 3: Alliance formation increases the probability of intervention: $\gamma(BC) > \gamma(\emptyset)$.

Proof: To prove this claim I consider each equilibrium separately. In equilibrium E0 no alliances form. However, by assumption $\gamma(BC) = 1 > y_c - x_c = \gamma(\emptyset)$. In equilibrium E1, $\gamma(BC) = 1 > \gamma(\emptyset) = y_c - x_c$. In equilibrium E2, $\gamma(BC) = 1 > \gamma(\emptyset) = 0$. In equilibrium E3, $\gamma(BC) = (z_c - x_c - \theta)/\theta > \gamma(\emptyset) = 0$. In equilibrium E4, alliances always form. However, by assumption $\gamma(\emptyset) = 0 < z_c - x_c + h_c = \gamma(BC)$. In equilibrium E2b, $\gamma(BC) = 1 > \rho/(\rho + 1 - \theta) = \gamma(\emptyset)$. For equilibrium E3b, suppose that the claim is not true. This implies that $\gamma(BC) \leqslant \gamma(\emptyset) \Rightarrow \beta(BC) \leqslant \beta(\emptyset) \Rightarrow \alpha(BC) \geqslant \alpha(\emptyset)$. If E3b is an equilibrium then $\text{Diff}(\theta) = \alpha(BC)[\beta(BC)[x_c - h_c] - 1] - \alpha(\emptyset)[\beta(\emptyset)x_c - 1] - f_c = 0$, where $\theta \in t_3$. Rearranging this expression yields $x_c(\alpha(BC)\beta(BC) - \alpha(\emptyset)\beta(\emptyset)) - \alpha(BC)\beta(BC)h_c - \alpha(BC) + \alpha(\emptyset) - f_c = 0$. Since $\alpha(BC) \geqslant \alpha(\emptyset)$, $(\alpha(BC)\beta(BC) - \alpha(\emptyset)\beta(\emptyset)) < 0$ and $f_c > 0$; $\text{Diff}(\theta) < 0$. This is a contradiction. Therefore, for all equilibria, alliance formation increases the probability of intervention.

Equilibrium Refinements

The standard problem with signaling games is that if in equilibrium all types send the same signal then what do you believe about the type of the sender if you see a signal that should not be sent in equilibrium. While this might seem a technicality since the signal will never be sent it is important in characterizing equilibria. The out of equilibrium beliefs of other players effect the incentives of a player to defect. Unfortunately, the sequential equilibrium concept does not tell us what to believe for out of equilibrium beliefs. There are several standard refinements (see Banks, 1991 p. 14–23 for a discussion of these and Nalebuff, 1991 for a discussion of these refinements in a deterrence situation). The intuition behind all of these refinements is to find those types that have the largest incentives to send the out of equilibrium messages and thus weight the out of equilibrium beliefs accordingly. The principle problem to using the standard refinements in this paper is that there are two receivers, *A* and *B*. There have been refinements used for multiple receivers, for example Gartner, Gibbons and Scharfstein (1988) and Farrell and Gibbons (1989) but these have not been used for continuous types and costly signalling. Rather than apply a standard refinement I will attempt to justify my modeling choices by similar arguments to those of Universal Divinity (Banks and Sobel, 1987).

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I assume that both players A and B have the same beliefs about the type that would send an out of equilibrium message. Given these beliefs they can calculate the probability of intervention should a war occur, call this probability λ. Consider E0, $\gamma(BC)=\lambda$ and $\gamma(\emptyset)=z_c-x_c$. Given these beliefs, $\alpha(l)$, $\beta(l)$, and $\text{Diff}(\cdot)=E[U_c(BC)]-E[U_c(\emptyset)]$ can be calculated. Now find the type of C that maximizes Diff$(\cdot)$ for any λ. For $w_c \in t_3 \cup t_2$ $\text{dDiff}(\cdot)/\text{d}w_c \leqslant 0$. For $w_c \in t_1$ $\text{dDiff}(\cdot)/\text{d}w_c = -\alpha(BC)\beta(BC)+\alpha(\emptyset)\beta(\emptyset)$ which can be $+$ or $-$. Thus either $w_c=0$ or $w_c=y_c-x_c$ has the greatest incentive to defect for any λ. Therefore, placing all the probability weight on these types implies that $\gamma(BC)=1$.

Now consider E4. $\gamma(BC)=z_c-x_c+h_c$ and $\gamma(\emptyset)=\lambda$. Now find the type of w_c that maximizes Diff$(\cdot)$ for any value of λ. For $w_c \in t_3$, $\text{dDiff}/\text{d}w_c=0$, for $w_c \in t_2$ $\text{dDiff}/\text{d}w_c = -\alpha(BC)\beta(BC)<0$ and for $w_c \in t_1$ $\text{dDiff}/\text{d}w_c = -\alpha(BC)\beta(BC)+\alpha(\emptyset)\beta(\emptyset)$. Thus $\text{argmax}_{wc}\ \text{Diff} \in \{0, t_3\}$. Given this result I assume that the type with the greatest incentive to defect is in t_3 and hence $\gamma(\emptyset)=0$. These assumptions are appealing when the equilibria where the types partially separate are considered because in all these cases $\gamma(BC)>\gamma(\emptyset)$.

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ACKNOWLEDGMENTS

I would like to thank David Austen-Smith, Jeff Banks, Bruce Bueno de Mesquita, David Kreps, Andrew Martin, Fiona McGillivray, Randy Siverson and several anonymous referees for their advise and comments. I am solely responsible for any remaining errors.

NOTES

1. In the model nation B has no option to decline the alliance. Given the payoff of the game, it is a dominant strategy for nation B to accept any commitment from nation C. Given this dominance, I do not consider B's role explicitly.
2. Consider a probit analysis of the decision to intervene. The theory suggest that conditional on A attacking the expected value of the stochastic error term is negative. B's decisions to retaliate introduces a Positive bias into the expected value of the stochastic error term.
3. The equilibria are described as types below θ forming alliances while those above θ do not. However all types $w_c \in t_3$ are indifferent between forming an alliance or not. Thus instead of considering the equilibrium E3 as described it could be considered as all types in t_3 are randomizing their strategy, such that $\forall w_c \in t_3$, $\sigma_c^* = \theta/(1-[z_c-x_c-h_c])$. In terms of the analysis the interpretation of a mixed strategy does not matter. Substantively the only relevant factor is that the proportion of types in t_3 forming an alliance is $(\theta-z_c-x_c+h_c)/(1-(z_c-x_c+h_c))$.

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CONTRIBUTOR

Alastair Smith is an assistant professor of political science at Washington University. He completed his Ph.D. in political science at the University of Rochester, NY. His research uses game theoretic models to generate empirical falsifiable hypotheses about international processes. He has written on the role of alliances in international conflict, the effect of domestic politics on foreign policy, international negotiations, endogenous tariff theory, international sanctions and endogenous election timing.

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